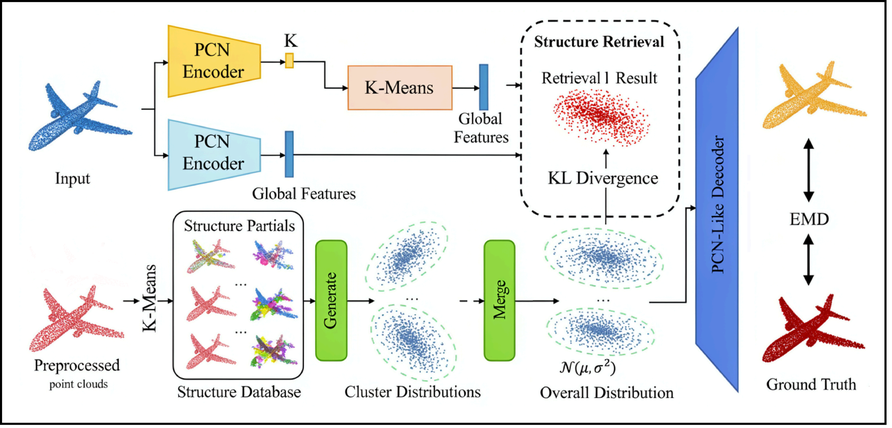
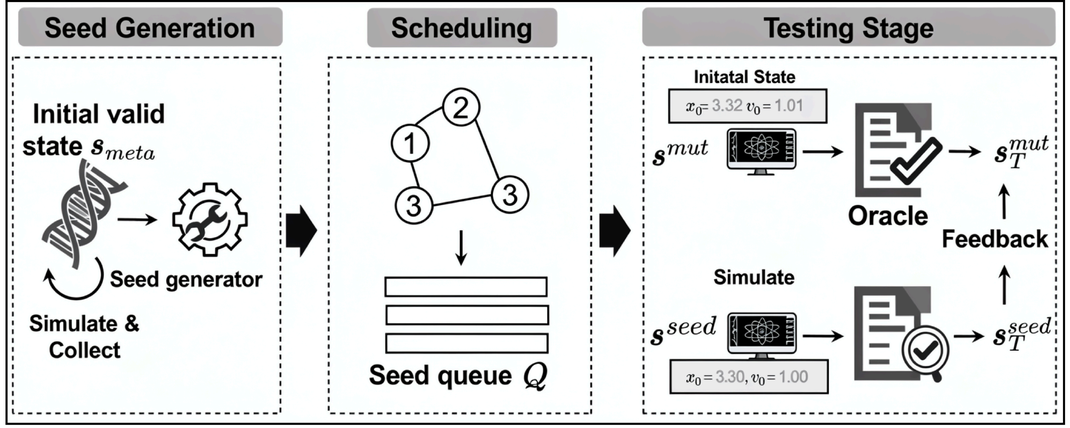
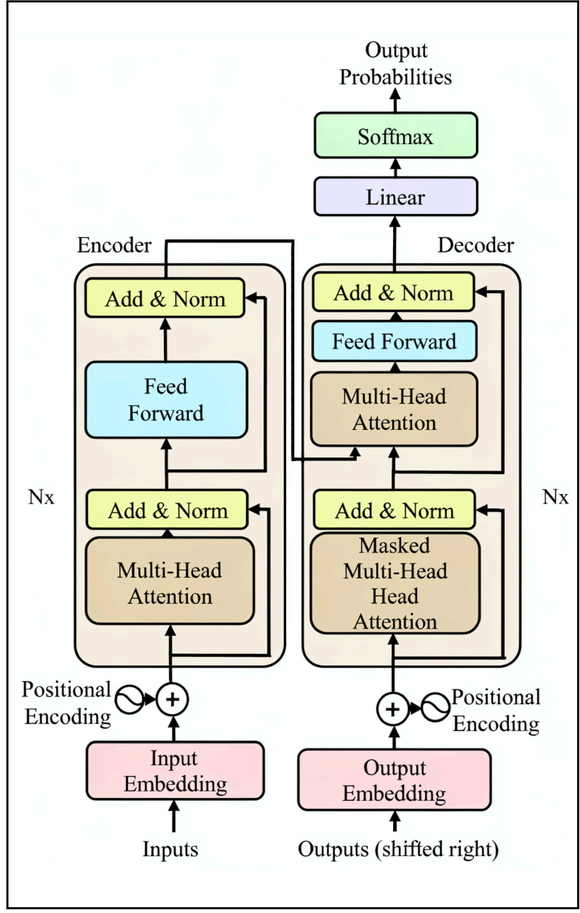
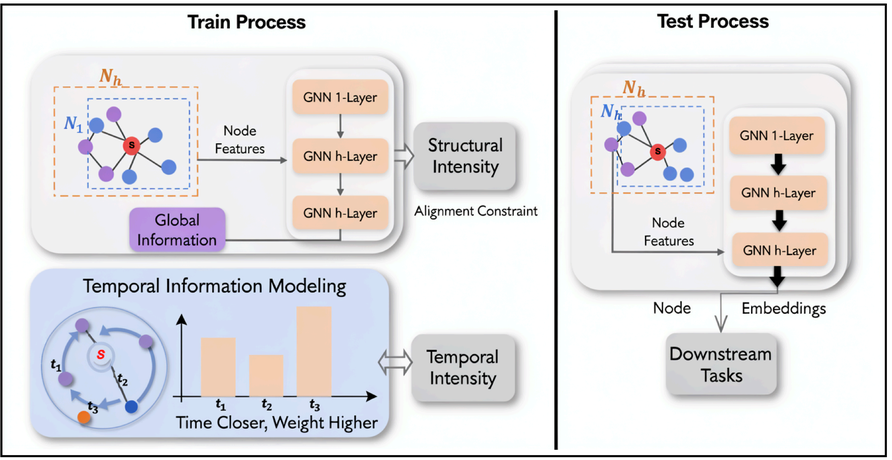
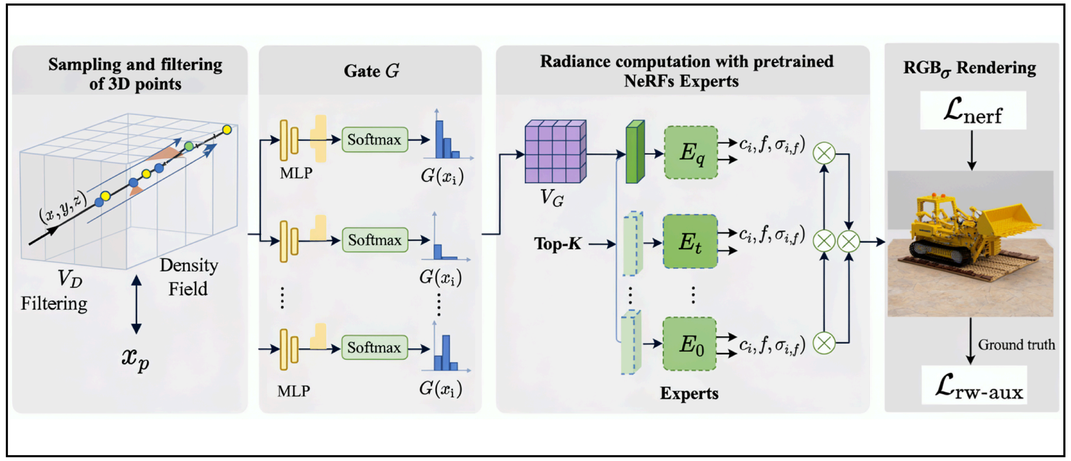
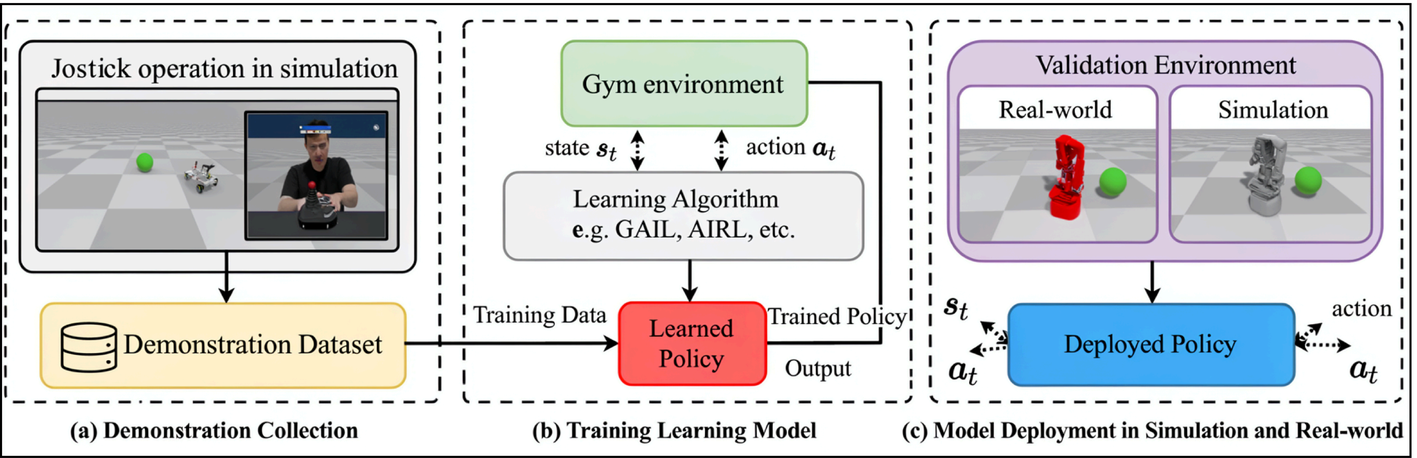

PCN Encoder
K
K-Means
Global Features
Structure Retrieval
Retrieval 1 Result
KL Divergence
Input
PCN Encoder
Global Features
PCN-Like Decoder
EMD
Structure Partials
K-Means
Generate
Merge
Preprocessed point clouds
Structure Database
Cluster Distributions
$\mathcal{N}(\mu, \sigma^2)$
Overall Distribution
Ground Truth

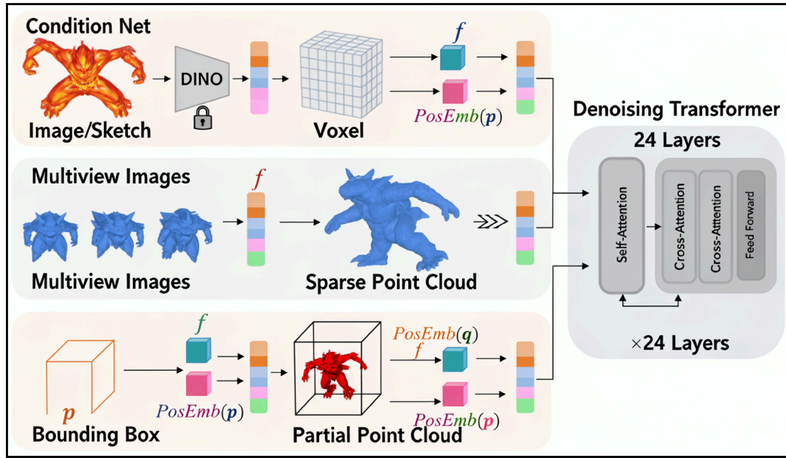

Seed Generation
Scheduling
Testing Stage
Initial valid state s_{meta}
Seed generator
Simulate & Collect
Seed queue Q
Initatal State
$x_0 = 3.32\ v_0 = 1.01$
s^{mut}
Oracle
s_T^{mut}
Feedback
Simulate
s^{seed}
$x_0 = 3.30, v_0 = 1.00$
s_T^{seed}


Output Probabilities
Softmax
Linear
Encoder
Decoder
Add & Norm
Feed Forward
Multi-Head Attention
Masked Multi-Head Head Attention
Nx
Positional Encoding
Input Embedding
Output Embedding
Inputs
Outputs (shifted right)


Train Process
Test Process
GNN 1-Layer
GNN h-Layer
Node Features
Structural Intensity
Alignment Constraint
Global Information
Temporal Information Modeling
Time Closer, Weight Higher
Temporal Intensity
Node Embeddings
Downstream Tasks


Sampling and filtering of 3D points
Gate G
Radiance computation with pretrained NeRFs Experts
RGB$_\sigma$ Rendering
Density Field
Filtering
MLP
Softmax
Top-K
Experts
$\mathcal{L}_{nerf}$
Ground truth
$\mathcal{L}_{rw\text{-}aux}$


Jostick operation in simulation
Demonstration Dataset
Gym environment
state s_t
action a_t
Learning Algorithm
e.g. GAIL, AIRL, etc.
Training Data
Learned Policy
Trained Policy
Output
Validation Environment
Real-world
Simulation
Deployed Policy
(a) Demonstration Collection
(b) Training Learning Model
(c) Model Deployment in Simulation and Real-world


Condition Net
DINO
Image/Sketch
Voxel
Multiview Images
Sparse Point Cloud
Bounding Box
Partial Point Cloud
PosEmb(p)
PosEmb(q)
Denoising Transformer
24 Layers
Self-Attention
Cross-Attention
Feed Forward
×24 Layers